



Quark mixing from muon collider neutrinos

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Outline

Common Ground

1. What is a Muon Collider
2. Neutrinos from Muon Collider
3. CKM, PDFs and FFs

Outline

Common Ground

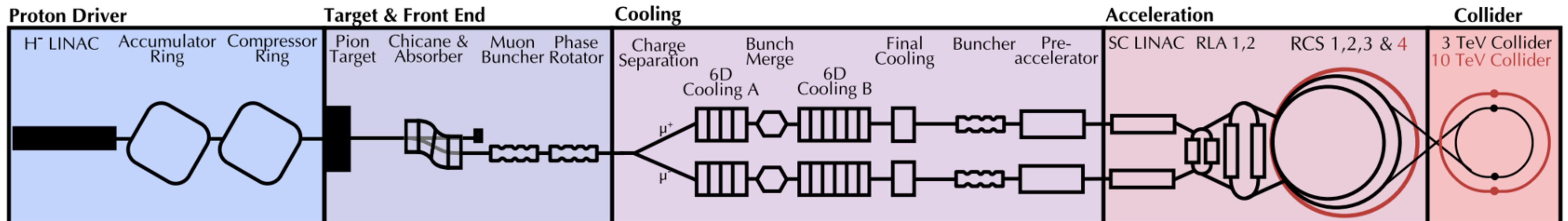
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Key Outcomes

4. ν DIS
5. CKM Determination
6. PDF Determination

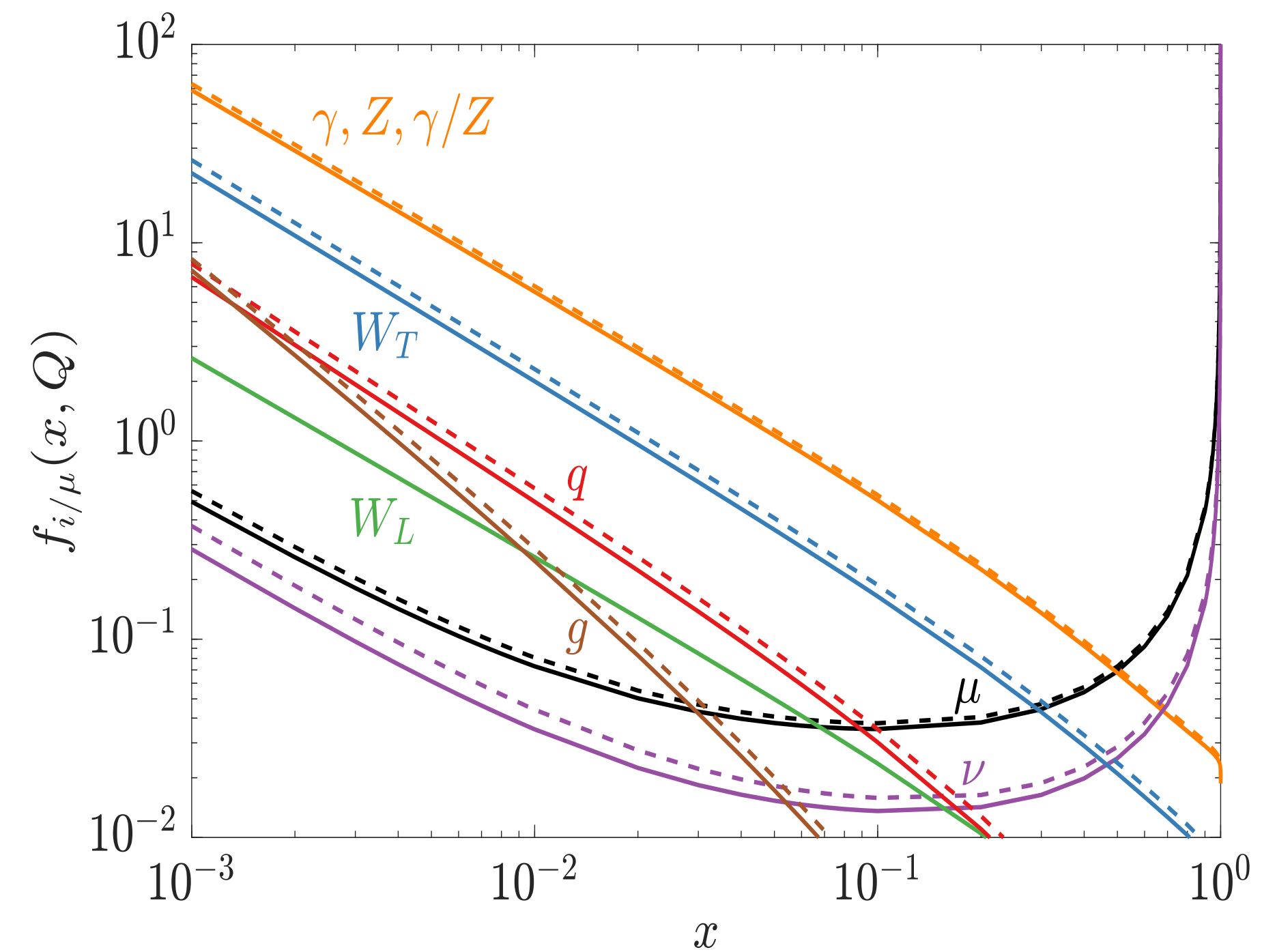
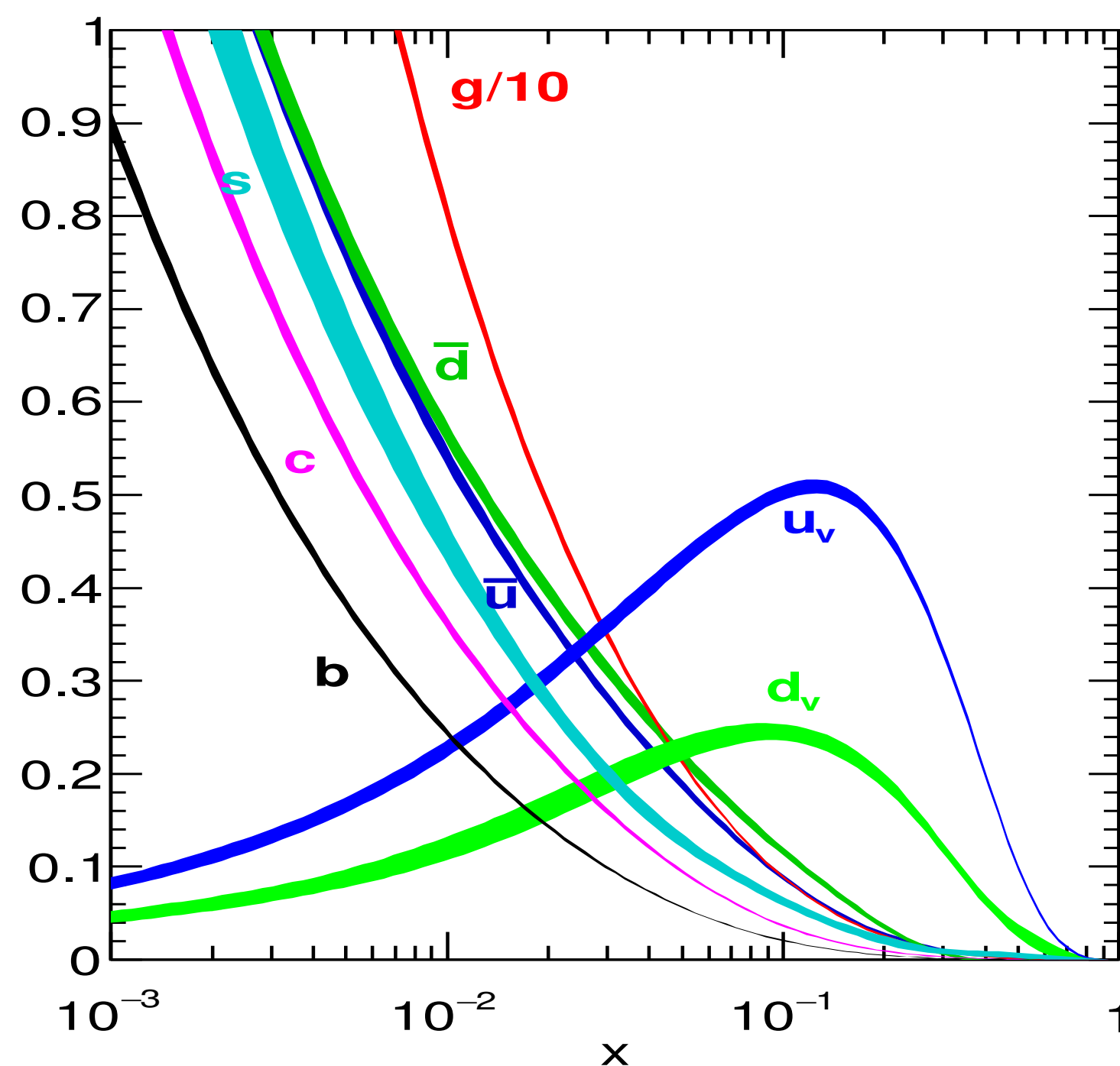
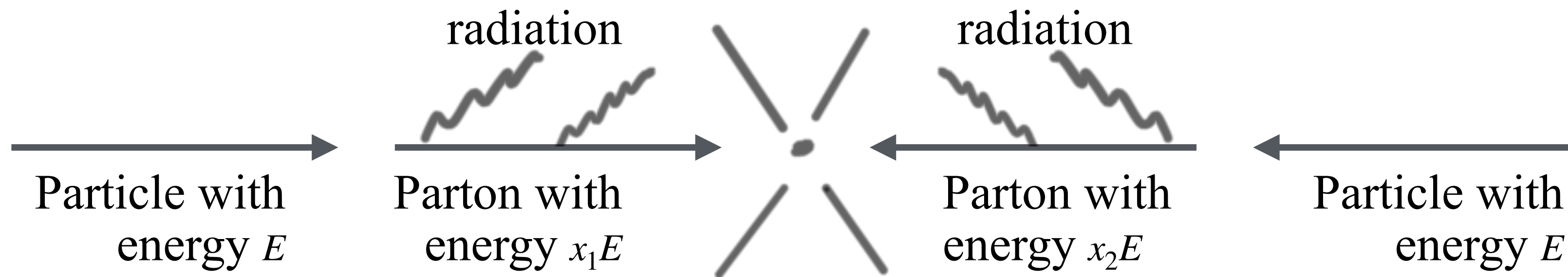
What is a Muon Collider?

- A hypothetical collider based on $\mu^+\mu^-$ collisions
- Muons are heavy $\rightarrow$ **negligible synchrotron radiation**
 $\rightarrow$ 10 TeV energy in a compact ring



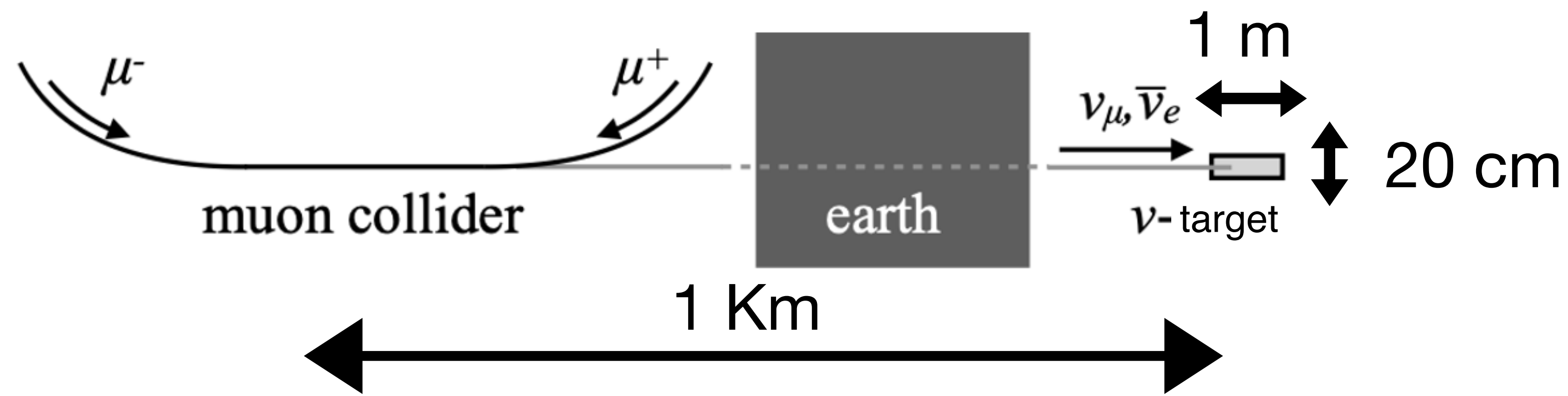
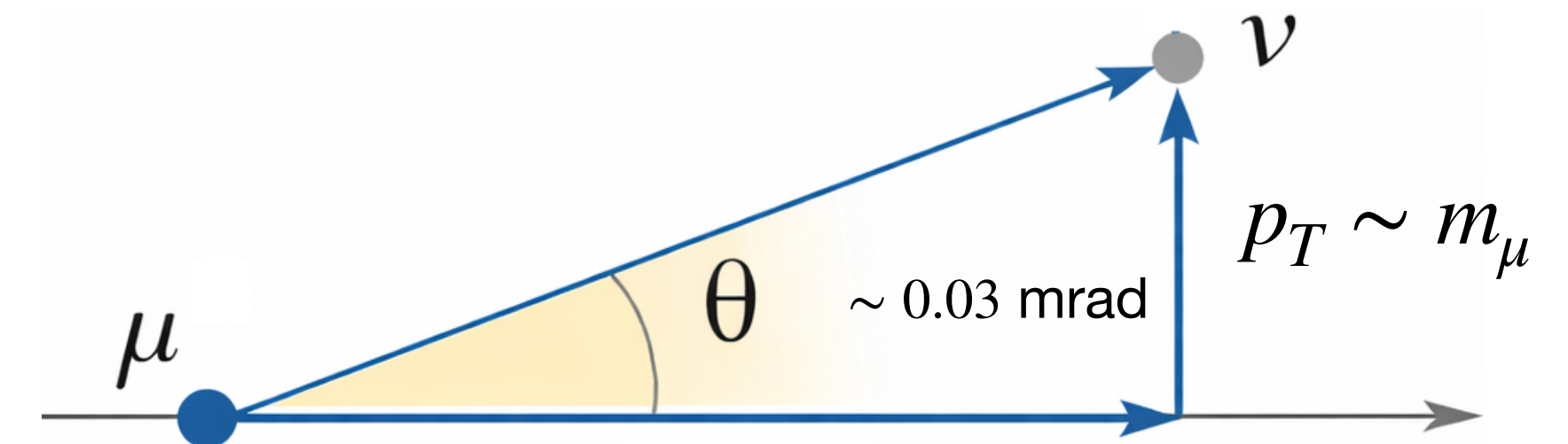
The Muon Collider - arXiv:2504.21417

10 TeV collisions with 10 TeV colliders



Neutrinos From Muon Collider

- Muons decay all along the collider ring, producing ν_μ and $\bar{\nu}_e$ with typical energies of **few TeV**
- Strong Lorentz boost $\rightarrow$ **highly collimated neutrino beam**
- Muon beam angular spread can broaden ν beam



Schematic of the ν MuC experimental setup

ν Rate at MuC

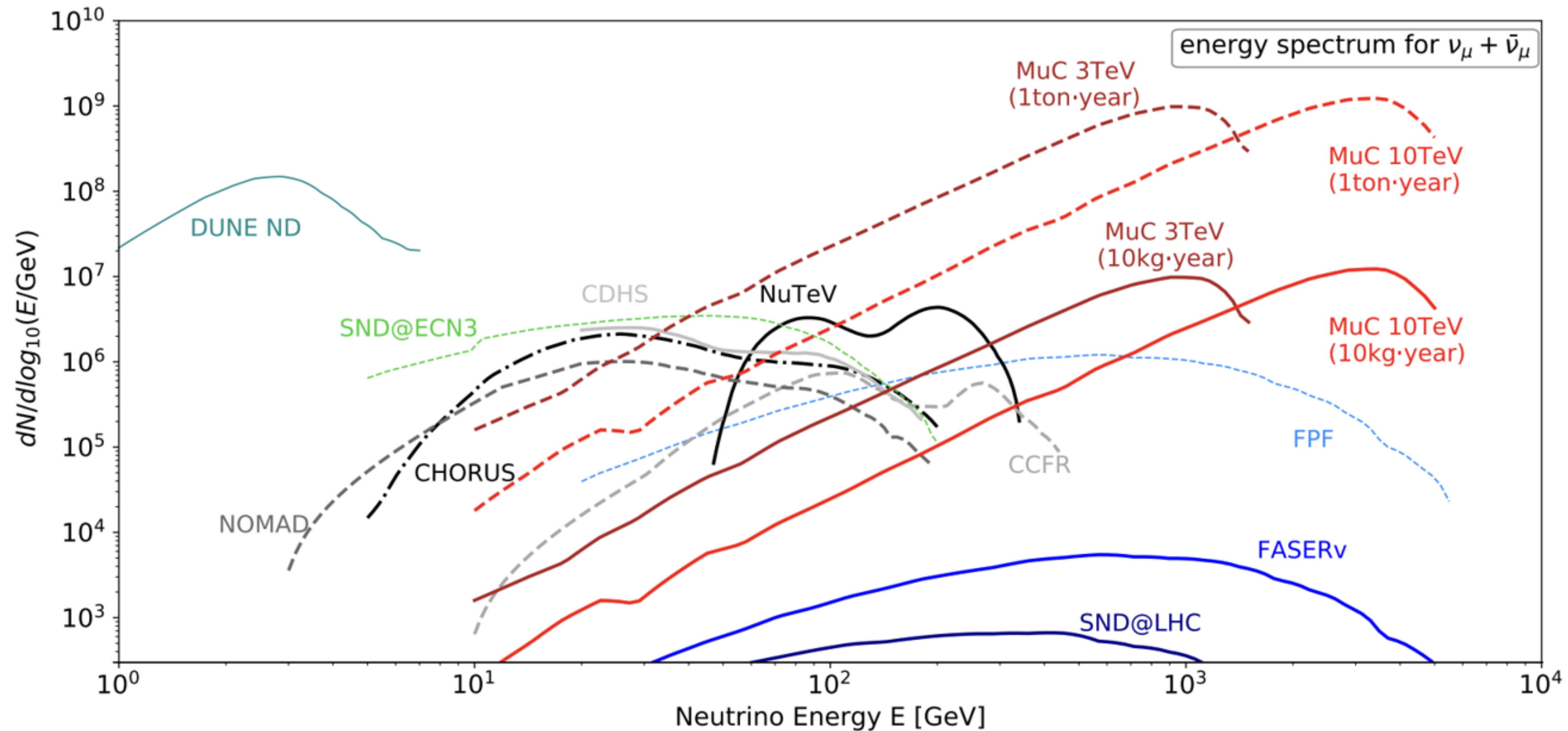
Parameter	Symbol	Unit	Scenario 1	
			Stage 1	Stage 2
Centre-of-mass energy	E_{cm}	TeV	3	10
Target integrated luminosity	$\int \mathcal{L}_{\text{target}}$	ab^{-1}	1	10
Estimated luminosity	$\mathcal{L}_{\text{estimated}}$	$10^{34} \text{cm}^{-2} \text{s}^{-1}$	2.1	21
Collider circumference	C_{coll}	km	4.5	10
Collider arc peak field	B_{arc}	T	11	16
Luminosity lifetime	N_{turn}	turns	1039	1558
Muons/bunch	N	10^{12}	2.2	1.8
Repetition rate	f_r	Hz	5	5
Beam power	P_{coll}	MW	5.3	14.4
RMS longitudinal emittance	$\varepsilon_{\parallel}$	eVs	0.025	0.025
Norm. RMS transverse emittance	$\varepsilon_{\perp}$	μm	25	25
IP bunch length	σ_z	mm	5	1.5
IP betafunction	β	mm	5	1.5
IP beam size	σ	μm	3	0.9
Protons on target/bunch	N_p	10^{14}	5	5
Proton energy on target	E_p	GeV	5	5

$$\mathcal{R}_{\nu} = N f_r \frac{L}{C_{\text{coll}}} = 10^{10} / \text{s}$$

Assuming **5 years** of data-taking,
1 ton cylindrical target
(r : 10 cm, L_{target} : 1 m):

$$\int \mathcal{L} = 1500 \text{ ab}^{-1}$$

ν Energy Spectrum at ν MuC



The energy spectrum of neutrino interactions produced in one year, for past and planned neutrino experiments. The solid and dashed lines assume, respectively, a small 10 kg and a realistic 1 ton target mass

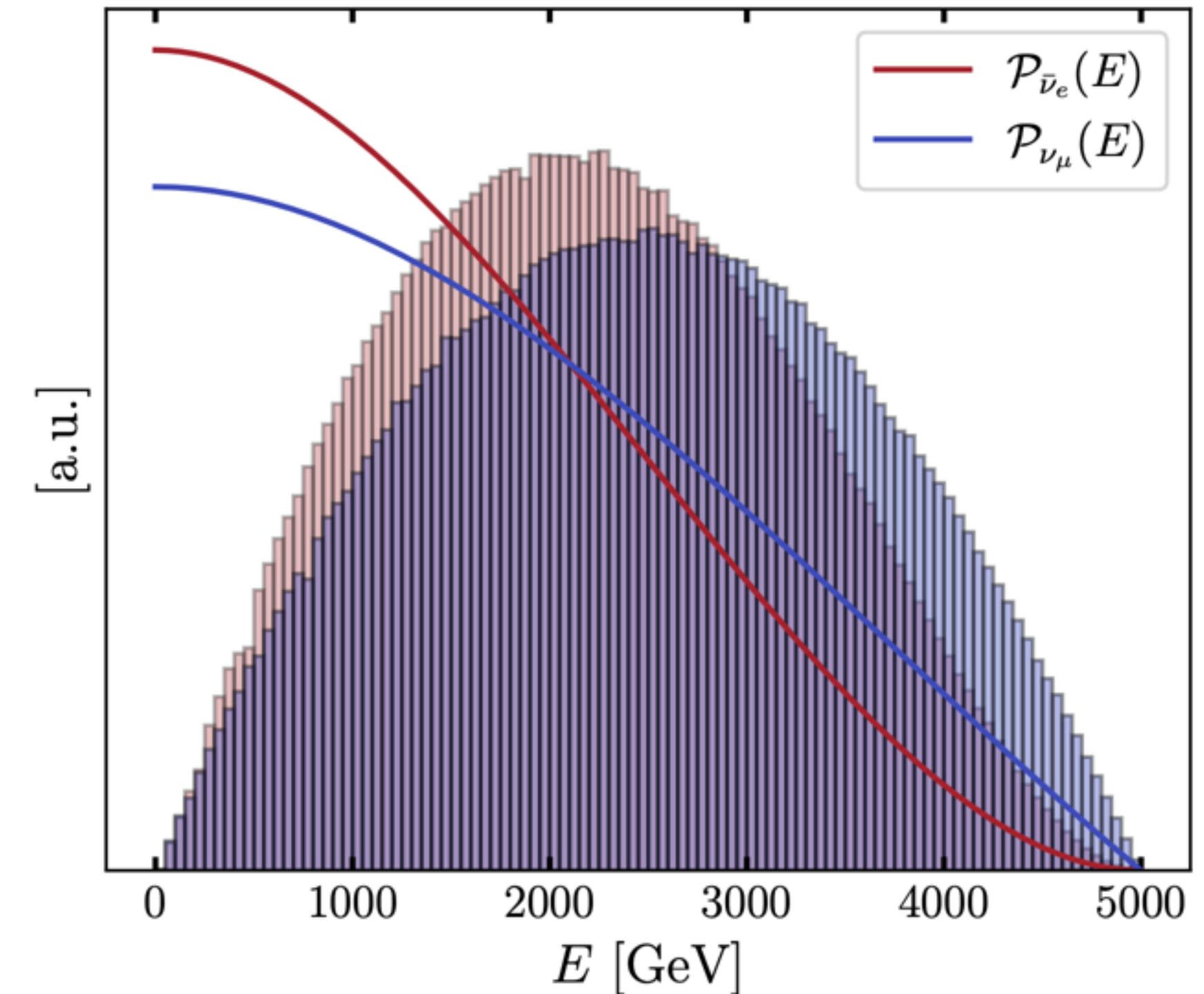
ν Energy Spectrum at ν MuC

Why is this special?

- A lot of neutrinos!
- Energy spectrum is **accurately predicted** unlike neutrinos from protons

$$\mathcal{P}_{\nu_\mu}(E) = \frac{1}{\Gamma} \frac{d\Gamma}{dE} = \frac{1}{3\bar{E}} \left[5 - 9 \left(\frac{E}{\bar{E}} \right)^2 + 4 \left(\frac{E}{\bar{E}} \right)^3 \right]$$

$$\beta = (1 - m_\mu^2/E_\mu^2)^{1/2} \approx 1, \quad \bar{E} = (1 + \beta)E_\mu/2 \approx E_\mu$$



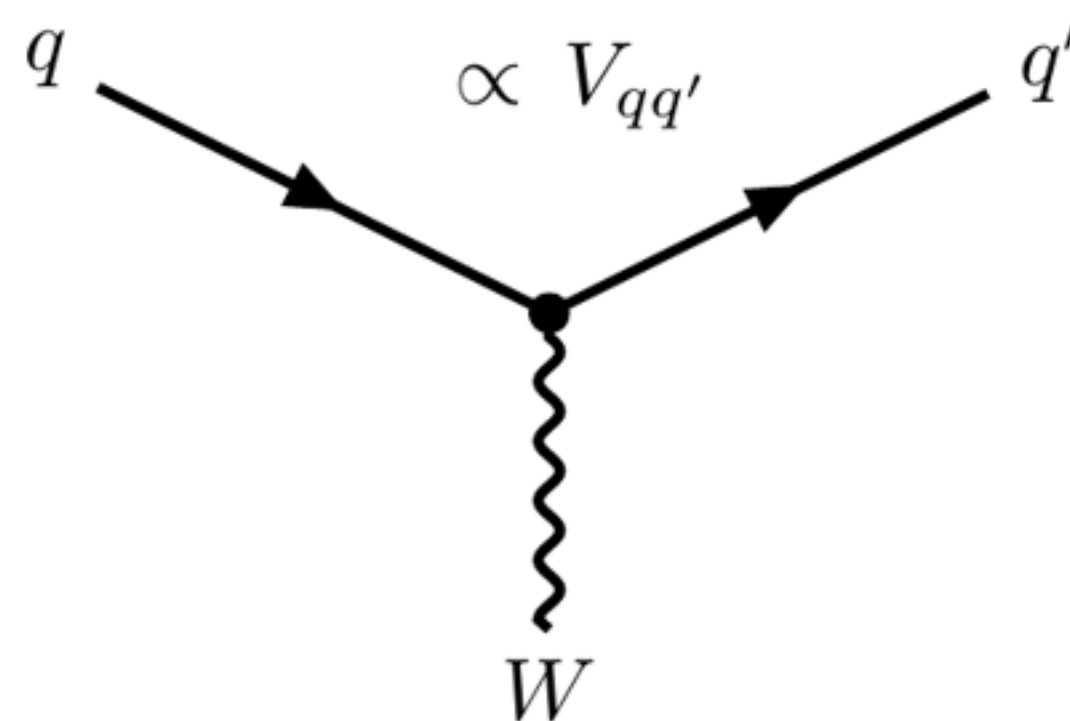
Energy distribution of the interacting neutrinos, compared with the neutrino beam energy spectrum

The Cabibbo-Kobayashi-Maskawa (CKM) Matrix

- Quark flavours mix under **weak interactions**
- CKM matrix encodes the strength of **charged-current transitions**
- In the SM it is a **unitary** 3×3 **matrix**
- Precise CKM tests probe the **consistency of the SM**

$$|V_{\text{CKM}}| = \begin{pmatrix} 0.974 & 0.225 & 0.004 \\ 0.225 & 0.973 & 0.042 \\ 0.009 & 0.041 & 0.999 \end{pmatrix}$$

PDG values (absolute magnitudes)

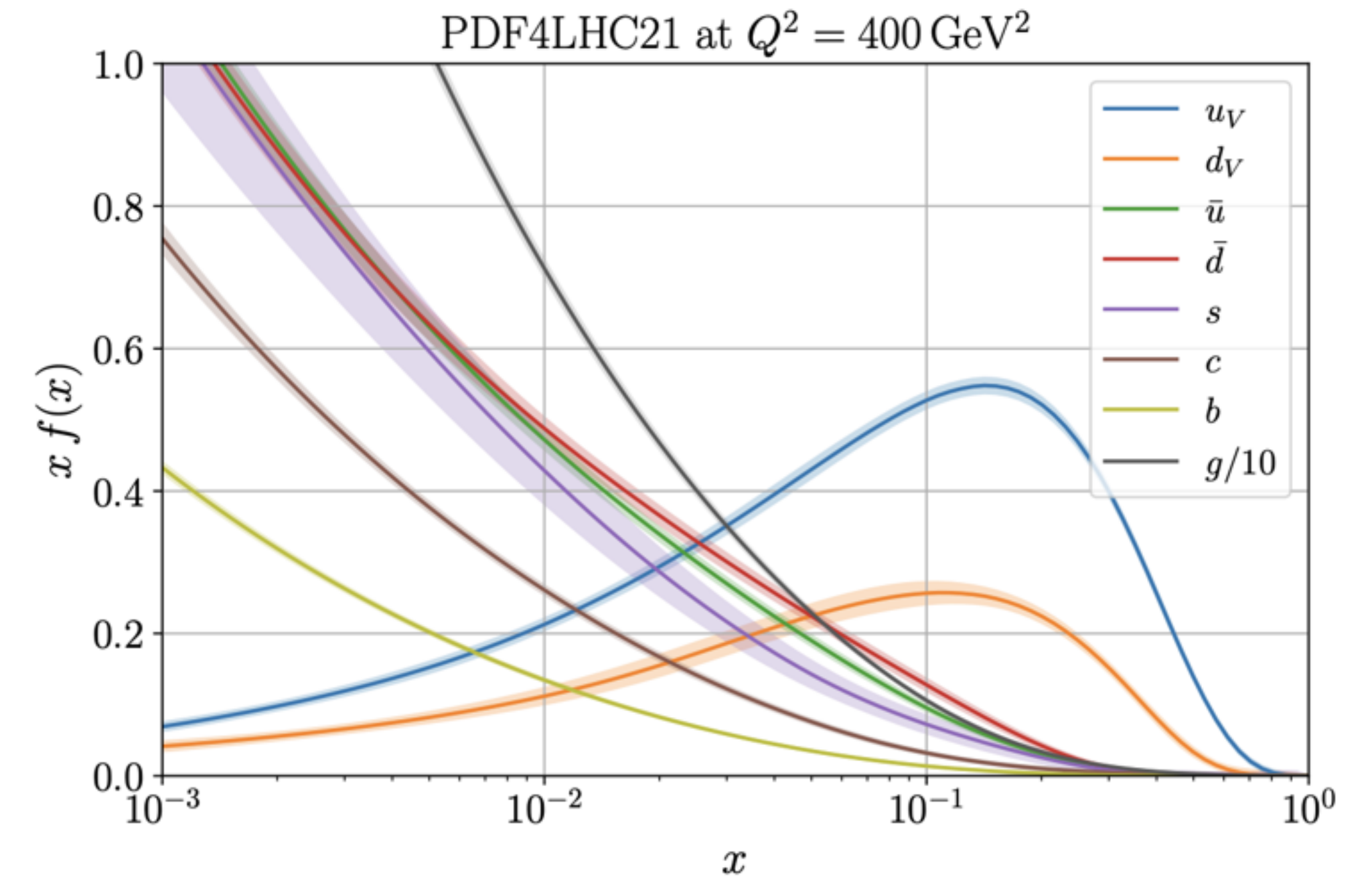


three generations of matter (fermions)		
I	II	III
$\simeq 2.2 \text{ MeV}$ $+\frac{2}{3}$ $\frac{1}{2}$ u up	$\simeq 1.3 \text{ GeV}$ $+\frac{2}{3}$ $\frac{1}{2}$ c charm	$\simeq 173 \text{ GeV}$ $+\frac{2}{3}$ $\frac{1}{2}$ t top
$\simeq 4.7 \text{ MeV}$ $-\frac{1}{3}$ $\frac{1}{2}$ d down	$\simeq 96 \text{ MeV}$ $-\frac{1}{3}$ $\frac{1}{2}$ s strange	$\simeq 4.2 \text{ GeV}$ $-\frac{1}{3}$ $\frac{1}{2}$ b bottom

Parton Distribution Functions (PDFs)

What are PDFs?

- Protons (and nucleons) are **not elementary particles**
They are made of **quarks and gluons (partons)**
- A PDF gives the probability to find a parton carrying a fraction x of the proton momentum
- The scale Q^2 sets the **resolution** of the probe
higher $Q^2 \rightarrow$ finer resolution inside the proton

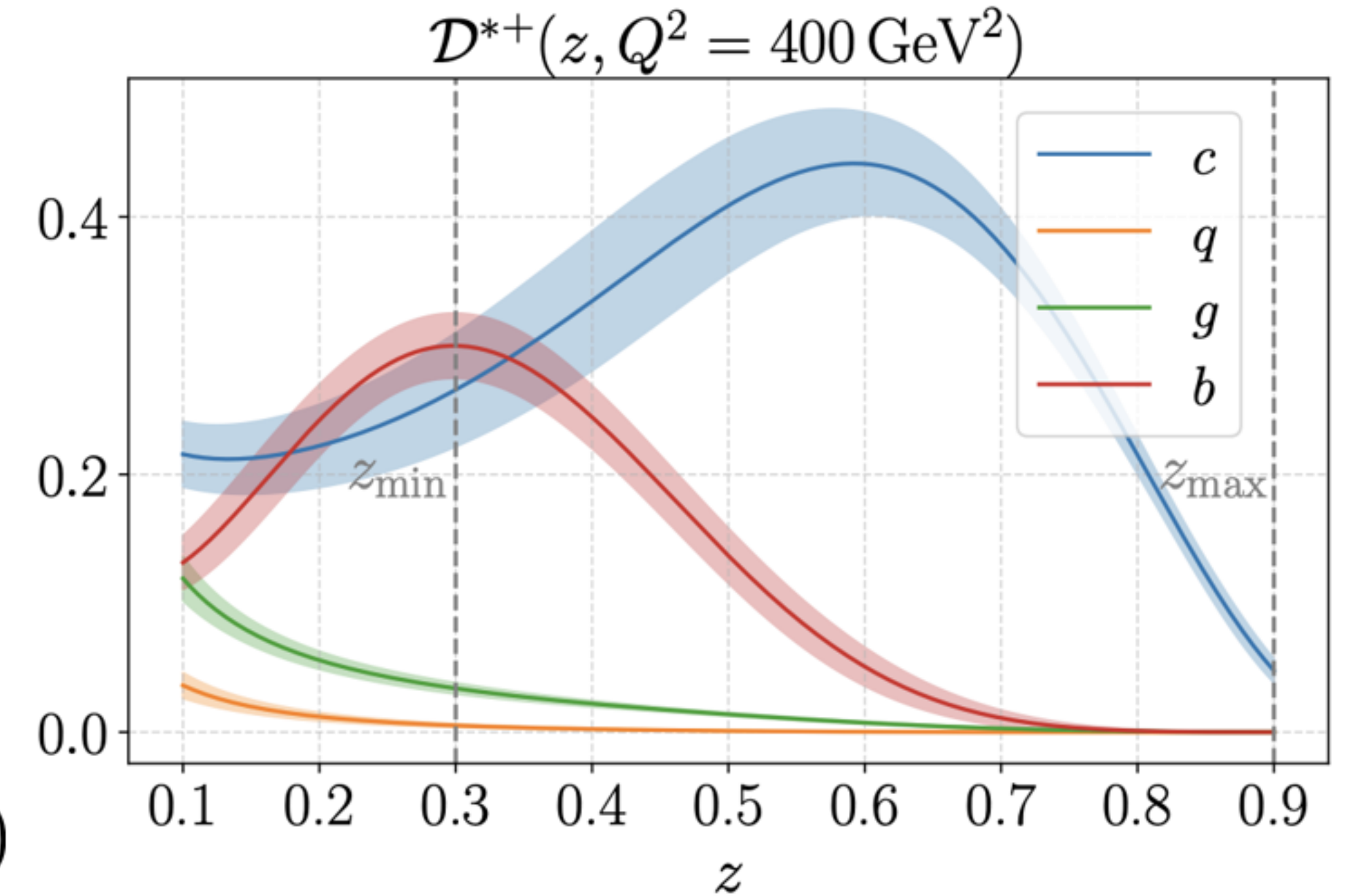


PDF4LHC21 for different channels at $Q^2 = 400 \text{ GeV}^2$

Fragmentation Functions (FFs)

Why FFs? (SIDIS)

- In **SIDIS** we measure a **tagged hadron** in the final state (e.g. $K^\pm$, $D^{*\pm}$, B -hadrons)
- FFs give the probability that a **parton** produces a given hadron carrying a fraction z of the parton energy
- The scale Q^2 controls the **evolution** of FFs (as for PDFs)
FFs are essential for **flavour tagging** in ν SIDIS



Deep Inelastic Scattering (DIS)

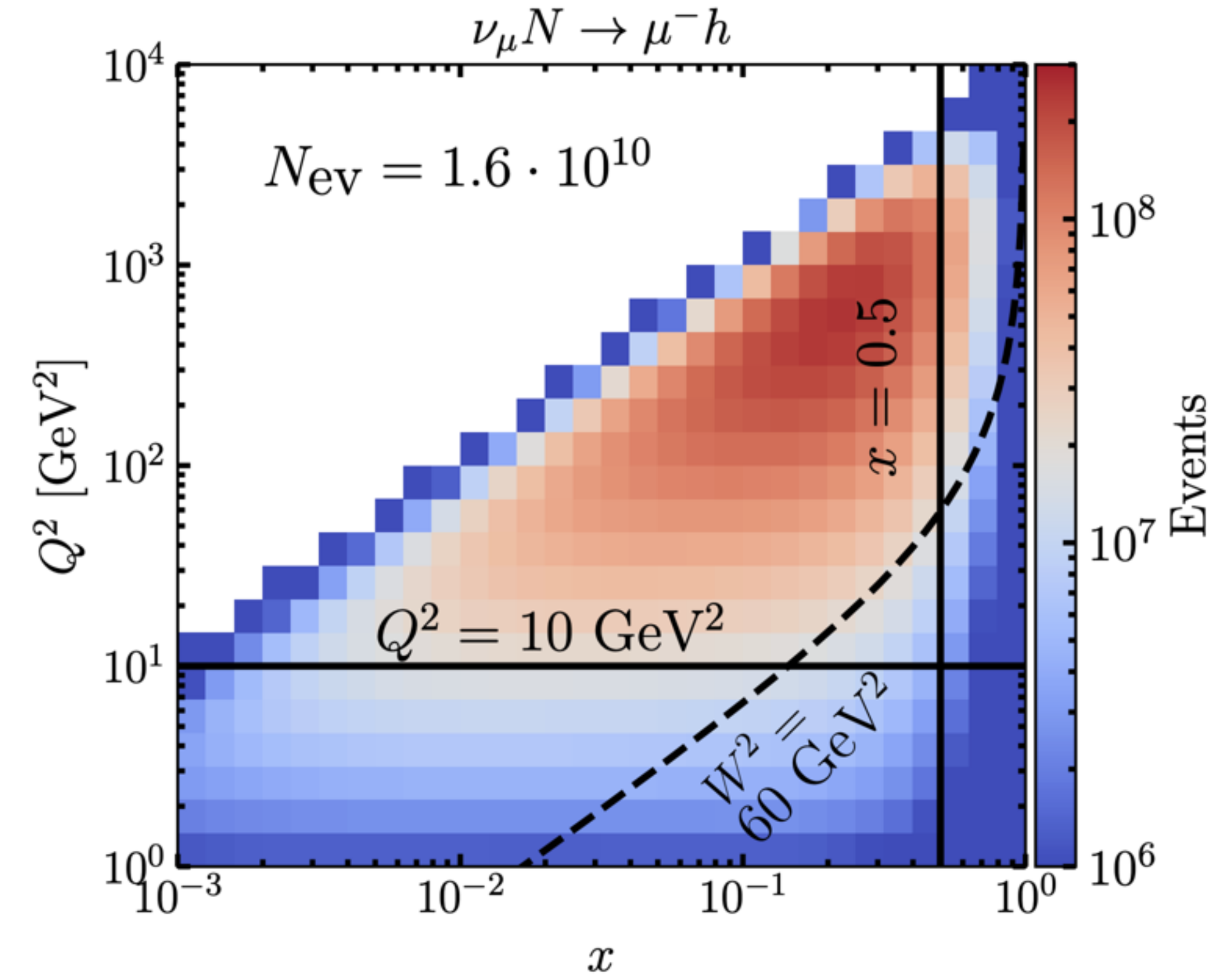
Neutrino–nucleon scattering

Charged Current (CC): $\nu_\mu + N \longrightarrow \mu^- + X$
 $\bar{\nu}_e + N \longrightarrow e^+ + X$

Neutral Current (NC): $\nu + N \longrightarrow \nu + X$

$$\frac{d^2\sigma}{dx dQ^2} \sim \sum_{q,q'} |V_{qq'}|^2 f_q(x, Q^2) \times \text{kin}(x, Q^2)$$

- DIS directly probes **quark flavour and momentum**



Number of events distribution in x - Q^2 bins

Deep Inelastic Scattering (DIS)

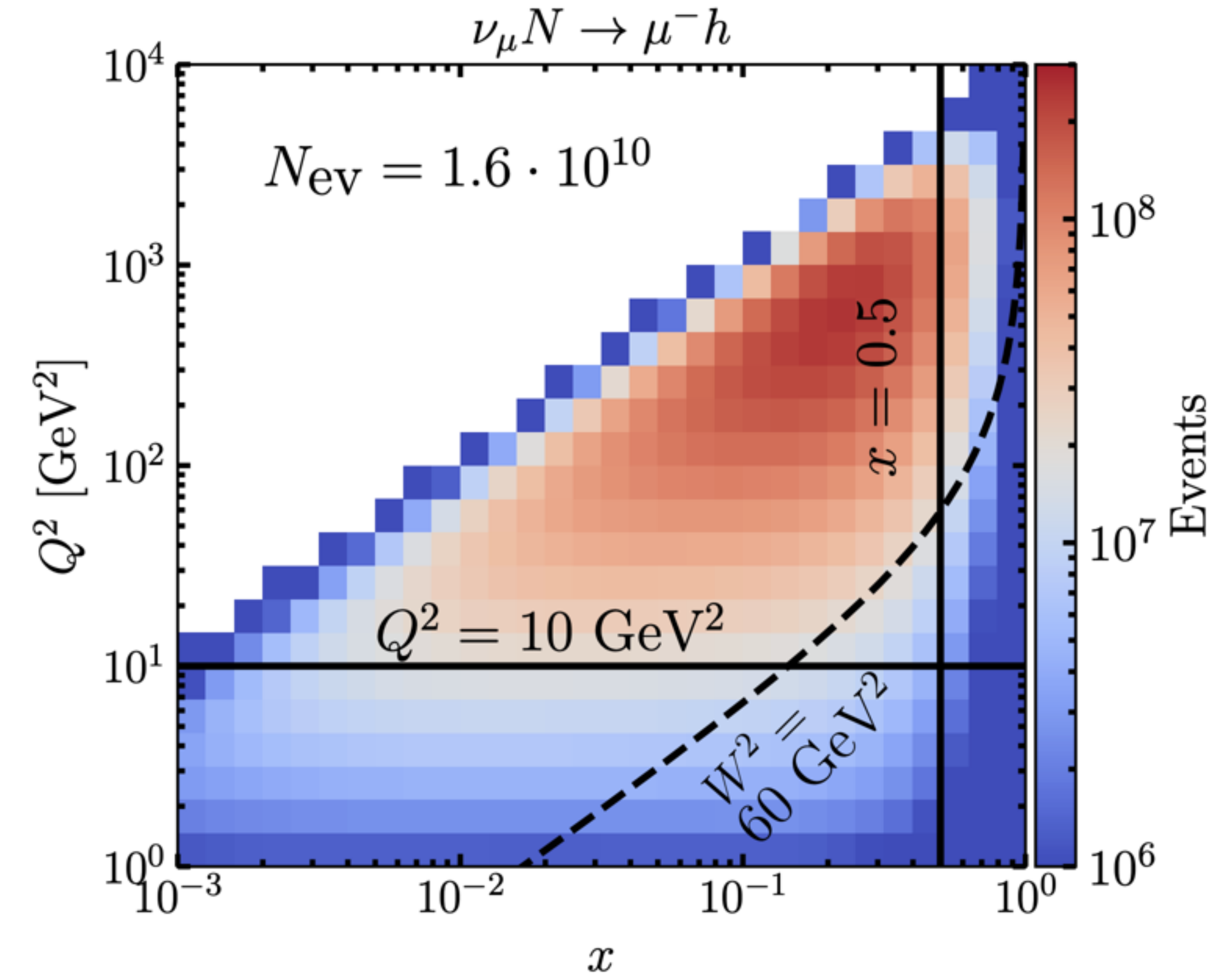
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- CKM elements control the **weak transition**



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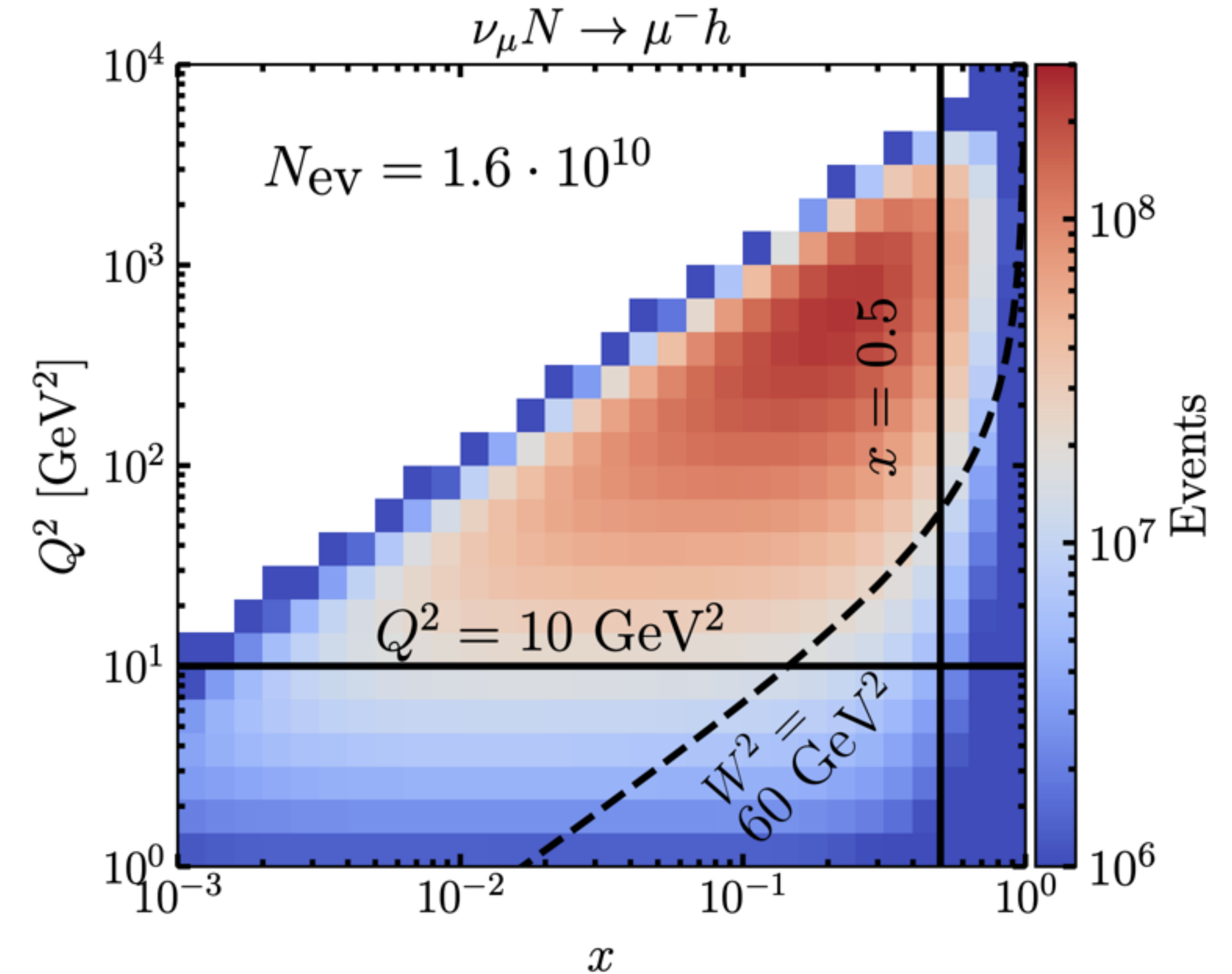
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$$\frac{d^2\sigma}{dx dQ^2} \sim \sum_{q,q'} |V_{qq'}|^2 \underbrace{f_q(x, Q^2)}_{\text{PDF}} \times \text{kin}(x, Q^2)$$

- DIS directly probes **quark flavour and momentum**
- CKM elements control the **weak transition**
- PDFs encode the **nucleon structure**



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Deep Inelastic Scattering (DIS)

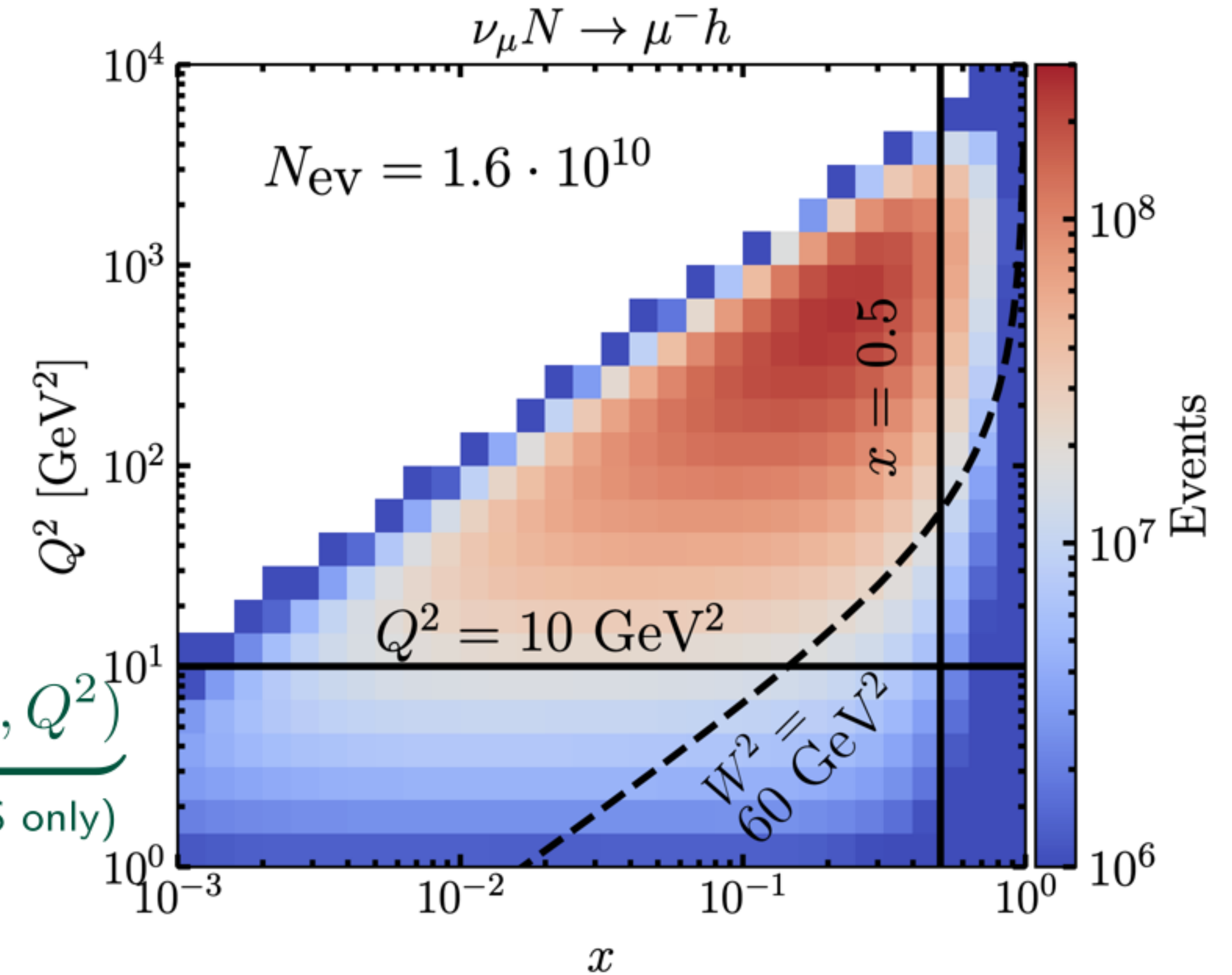
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$$\frac{d^2\sigma}{dx dQ^2} \sim \sum_{q,q'} |V_{qq'}|^2 f_q(x, Q^2) \times \text{kin}(x, Q^2) \times \underbrace{F_{h/q'}(z, Q^2)}_{\text{FF (SIDIS only)}}$$

- DIS directly probes **quark flavour and momentum**
- CKM elements control the **weak transition**
- PDFs encode the **nucleon structure**



Number of events distribution in x – Q^2 bins

Number of DIS Events at ν MuC

	Inclusive		s -tagged		$\bar{s}$ -tagged	
	Process	N_{ev}	Process	N_{ev}	Process	N_{ev}
CC	$\nu_\mu W \rightarrow \mu^- h$	$1.5 \cdot 10^{10}$	$\nu_\mu W \rightarrow \mu^- K^-$	$2.4 \cdot 10^8$	$\nu_\mu W \rightarrow \mu^- K^+$	$7.7 \cdot 10^8$
	$\bar{\nu}_e W \rightarrow e^+ h$	$6.4 \cdot 10^9$	$\bar{\nu}_e W \rightarrow e^+ K^-$	$2.9 \cdot 10^8$	$\bar{\nu}_e W \rightarrow e^+ K^+$	$1.5 \cdot 10^8$
NC	$\nu_\mu W \rightarrow \nu_\mu h$ $\bar{\nu}_e W \rightarrow \bar{\nu}_e h$	$7.5 \cdot 10^9$	$\nu_\mu W \rightarrow \nu_\mu K^-$ $\bar{\nu}_e W \rightarrow \bar{\nu}_e K^-$	$1.9 \cdot 10^8$	$\nu_\mu W \rightarrow \nu_\mu K^+$ $\bar{\nu}_e W \rightarrow \bar{\nu}_e K^+$	$2.5 \cdot 10^8$
	c -tagged		$\bar{c}$ -tagged		bottom-tagged	
	Process	N_{ev}	Process	N_{ev}	Process	N_{ev}
CC	$\nu_\mu W \rightarrow \mu^- D^{*+}$	$4.4 \cdot 10^8$	$\nu_\mu W \rightarrow \mu^- D^{*-}$	$8.0 \cdot 10^6$	$\nu_\mu W \rightarrow \mu^- B$	$1.7 \cdot 10^6$
	$\bar{\nu}_e W \rightarrow e^+ D^{*+}$	$3.3 \cdot 10^6$	$\bar{\nu}_e W \rightarrow e^+ D^{*-}$	$2.6 \cdot 10^8$	$\bar{\nu}_e W \rightarrow e^+ B$	$8.3 \cdot 10^5$
NC	$\nu_\mu W \rightarrow \nu_\mu D^{*+}$ $\bar{\nu}_e W \rightarrow \bar{\nu}_e D^{*+}$	$3.7 \cdot 10^7$	$\nu_\mu W \rightarrow \nu_\mu D^{*-}$ $\bar{\nu}_e W \rightarrow \bar{\nu}_e D^{*-}$	$3.5 \cdot 10^7$	$\nu_\mu W \rightarrow \nu_\mu B$ $\bar{\nu}_e W \rightarrow \bar{\nu}_e B$	$5.2 \cdot 10^7$

The DIS and SIDIS processes with the corresponding total number of expected events with a tungsten nucleon (denoted as W)

Observables

	Observable	Binning
CC	$R_{ijk}^{\mu/e} = \frac{\sigma_{ijk}^{\mu h}}{\sigma_{ijk}^{eh}}$ $R_{ijk}^{eb} = \frac{\sigma_{ijk}^{eB}}{\sigma_{ijk}^{eh}} \quad R_{ijk}^{\mu b} = \frac{\sigma_{ijk}^{\mu B}}{\sigma_{ijk}^{\mu h}} \quad R_{ijk}^{e\bar{c}} = \frac{\sigma_{ijk}^{eD^{*-}}}{\sigma_{ijk}^{eh}} \quad R_{ijk}^{\mu c} = \frac{\sigma_{ijk}^{\mu D^{*+}}}{\sigma_{ijk}^{\mu h}}$ $R_{ijk}^{es} = \frac{\sigma_{ijk}^{eK^-}}{\sigma_{ijk}^{eh}} \quad R_{ijk}^{\mu s} = \frac{\sigma_{ijk}^{\mu K^-}}{\sigma_{ijk}^{\mu h}} \quad R_{ijk}^{e\bar{s}} = \frac{\sigma_{ijk}^{eK^+}}{\sigma_{ijk}^{eh}} \quad R_{ijk}^{\mu \bar{s}} = \frac{\sigma_{ijk}^{\mu K^+}}{\sigma_{ijk}^{\mu h}}$	(x, Q^2, E) $27 \times 24 \times 5$
NC	$R_{ijk}^{CC/NC} = \sum_k \frac{\sigma_{ijk}^{eh} + \sigma_{ijk}^{\mu h}}{\sigma_{ij}^{\nu h}}$ $R_{ij}^{\nu b} = \frac{\sigma_{ij}^{\nu B}}{\sigma_{ij}^{\nu h}} \quad R_{ij}^{\nu c} = \frac{\sigma_{ij}^{\nu D^{*+}}}{\sigma_{ij}^{\nu h}} \quad R_{ij}^{\nu \bar{c}} = \frac{\sigma_{ij}^{\nu D^{*-}}}{\sigma_{ij}^{\nu h}}$ $R_{ij}^{\nu s} = \frac{\sigma_{ij}^{\nu K^-}}{\sigma_{ij}^{\nu h}} \quad R_{ij}^{\nu \bar{s}} = \frac{\sigma_{ij}^{\nu K^+}}{\sigma_{ij}^{\nu h}}$	(x, Q^2) 27×24

The observables employed in the fit

Parametric Uncertainties

- Parameters of interest: **CKM elements** and $\sin \theta_W$
Observables depend on **PDFs** and **Fragmentation Functions**
- PDF and FF uncertainties are included via **nuisance parameters**
Hessian representations of the PDFs and of the FFs $\rightarrow$

$$f_{\alpha}(\xi, Q^2; \nu) = f_{\alpha}^{(0)}(\xi, Q^2) + \sum_{k=1}^N \nu_k \left[f_{\alpha}^{(k)}(\xi, Q^2) - f_{\alpha}^{(0)}(\xi, Q^2) \right]$$

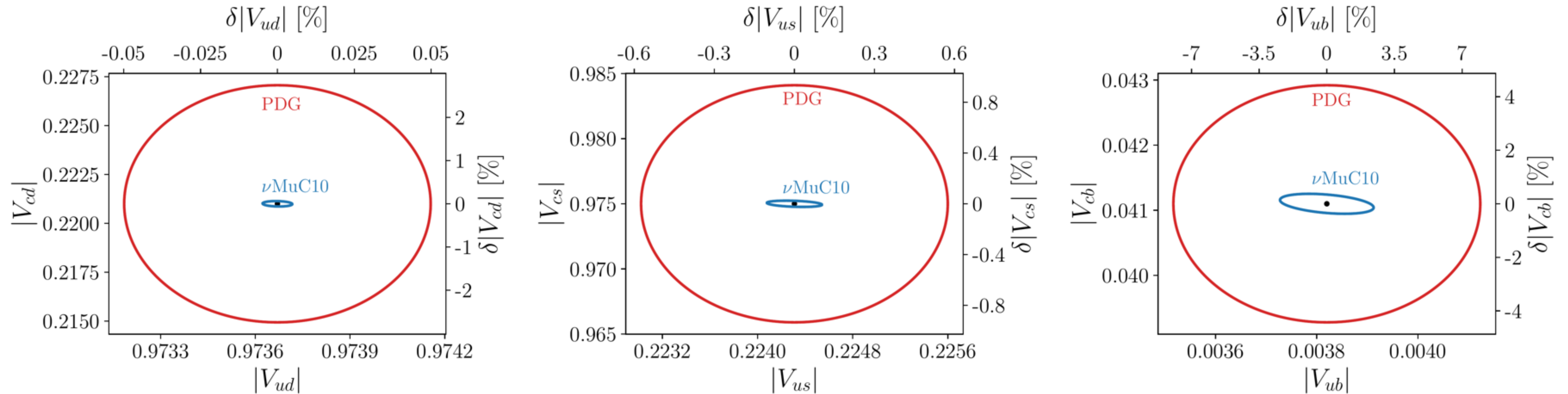
- Predictions are **linearised** in nuisance parameters
 $\rightarrow$ likelihood is quadratic and can be **profiled analytically**

CKM matrix elements determination

Relative Precision [%]	PDG	CC	(CC+NC) _{Stat}	νMuC_{10} (CC+NC)	Gain PDG/ νMuC_{10}
$ V_{ud} $	0.033	0.032	0.0019	0.0032	10
$ V_{us} $	0.38	0.19	0.017	0.067	5.6
$ V_{cd} $	1.8	1.6	0.0082	0.041	44
$ V_{cs} $	0.62	0.24	0.0033	0.016	39
$ V_{cb} $	2.9	0.63	0.19	0.24	12
$ V_{ub} $	5.2	1.6	1.5	1.6	3.3
$\sin \theta_W$	0.017	—	0.0095	0.015	1.1

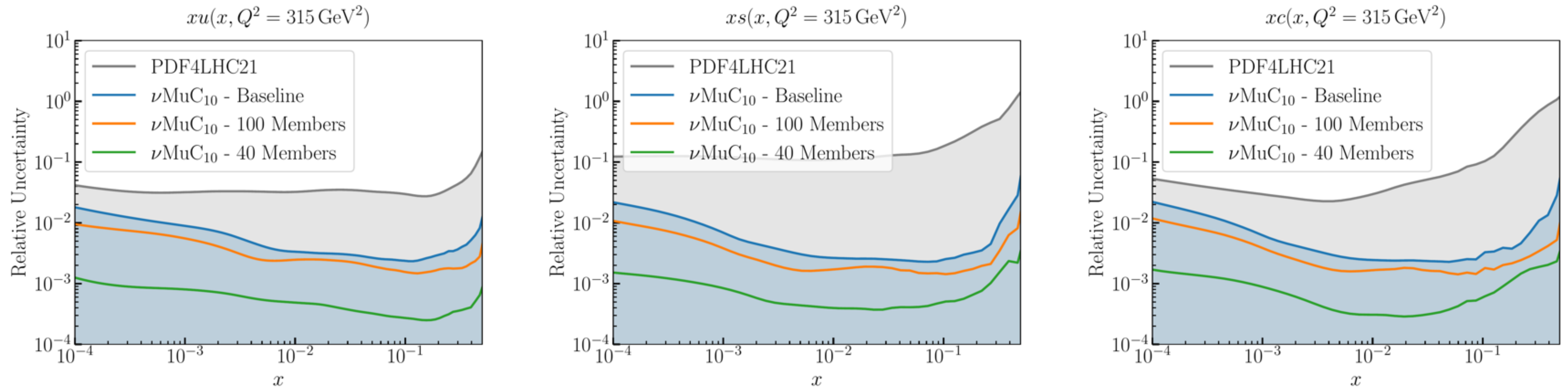
Relative global precision, in %, on the absolute values of CKM elements and on $\sin \theta_W$ from the PDG and after adding measurements at the 10 TeV νMuC in the baseline fit configuration

CKM matrix elements determination



68% CL ellipses for CKM pairs, showing the PDG prior (red)
and the baseline fit (blue)

PDF determination



PDF uncertainties expected at the νMuC experiment, compared with current uncertainties from PDF4LHC21

Conclusions

- A **parasitic neutrino experiment** at a muon collider (ν MuC) enables a **radical improvement** in CKM determination
- Novel and **independent** methodology from low-energy measurements based on a global analysis of **DIS and SIDIS**
- A strong improvement in **PDF determination** emerges as a byproduct
- Excellent performance already at **3 TeV**, with best reach at **10 TeV**

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Future directions

- Assessment of QCD uncertainties
- PDF Fit from scratch

Conclusions

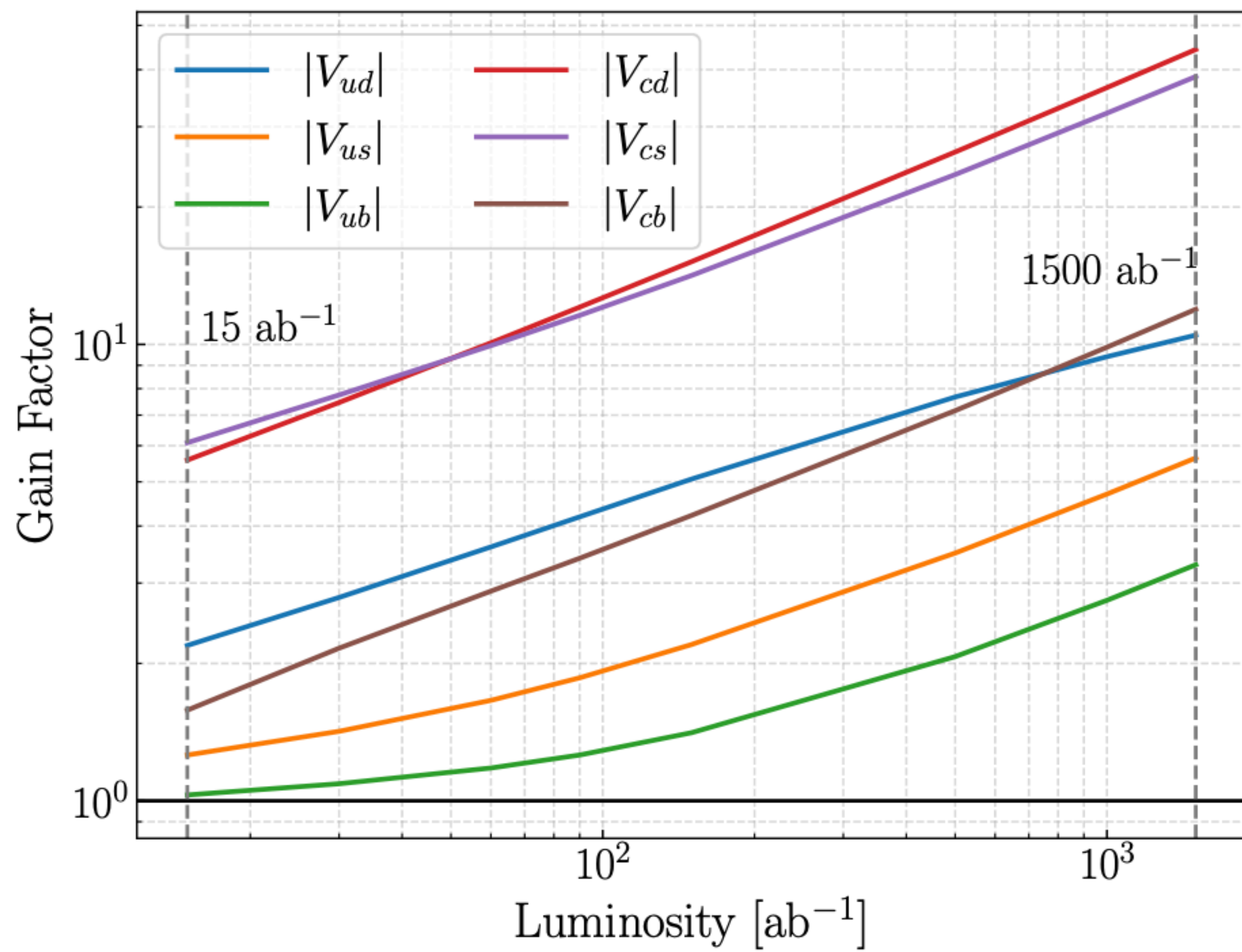
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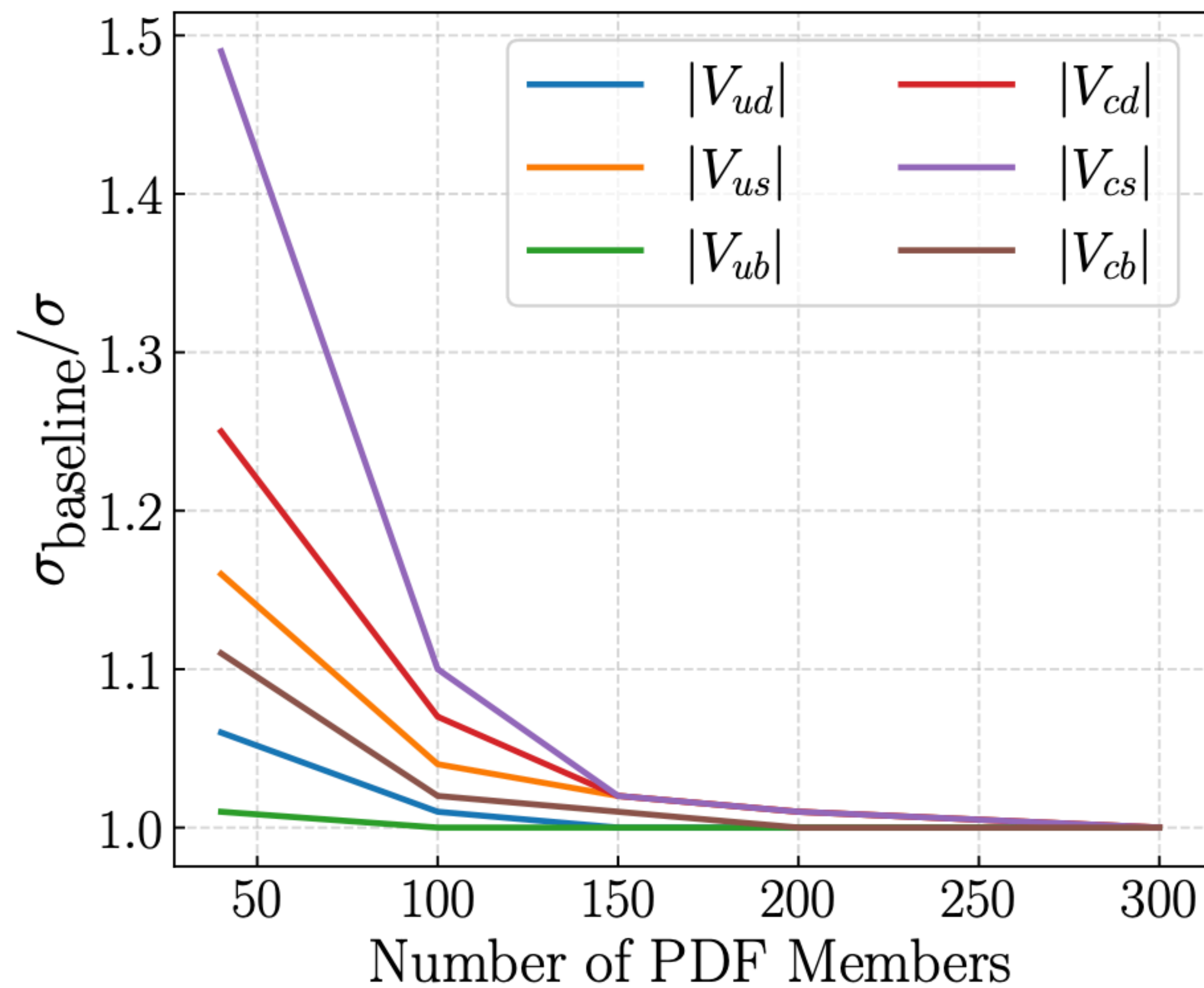
Future directions

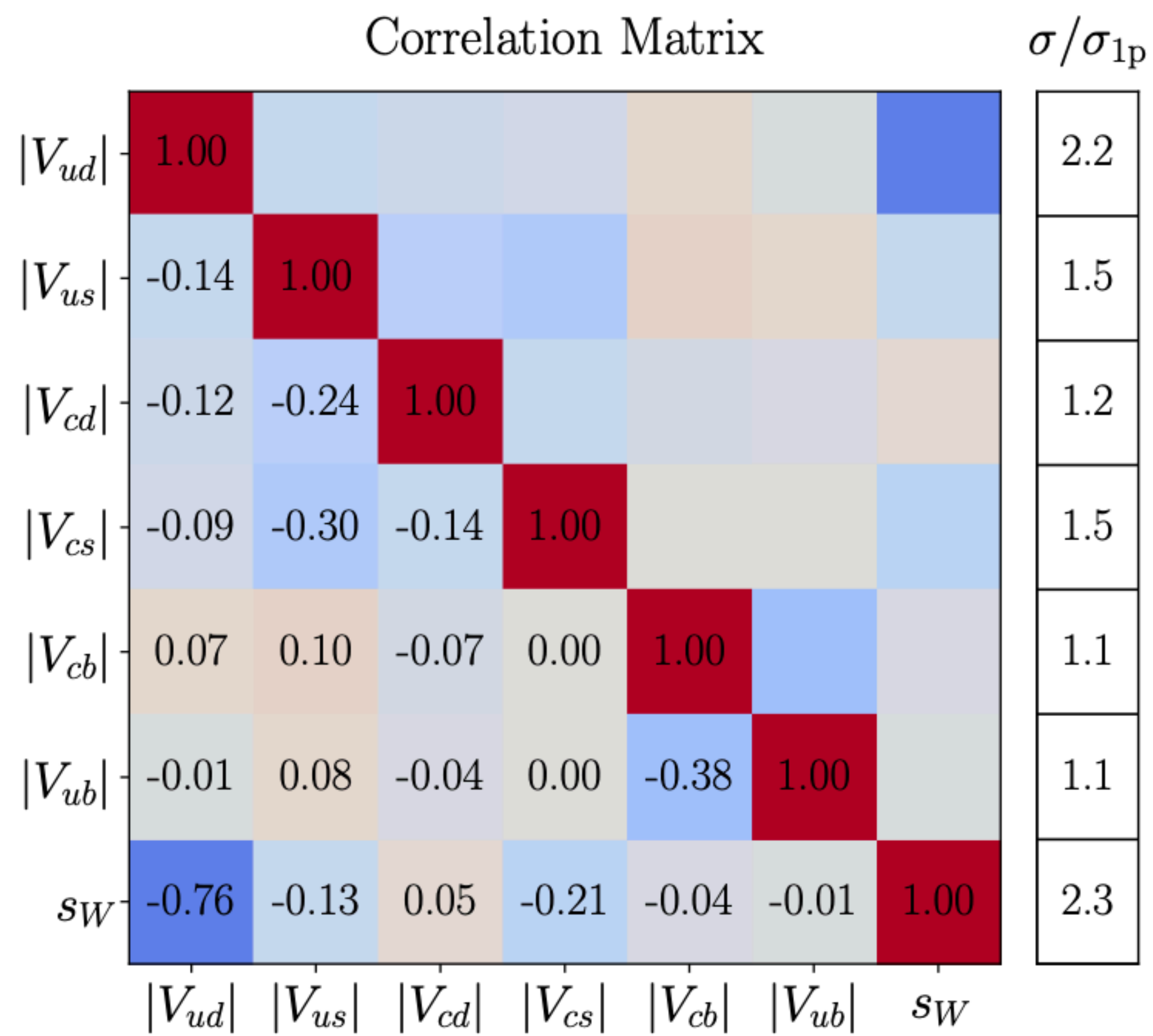
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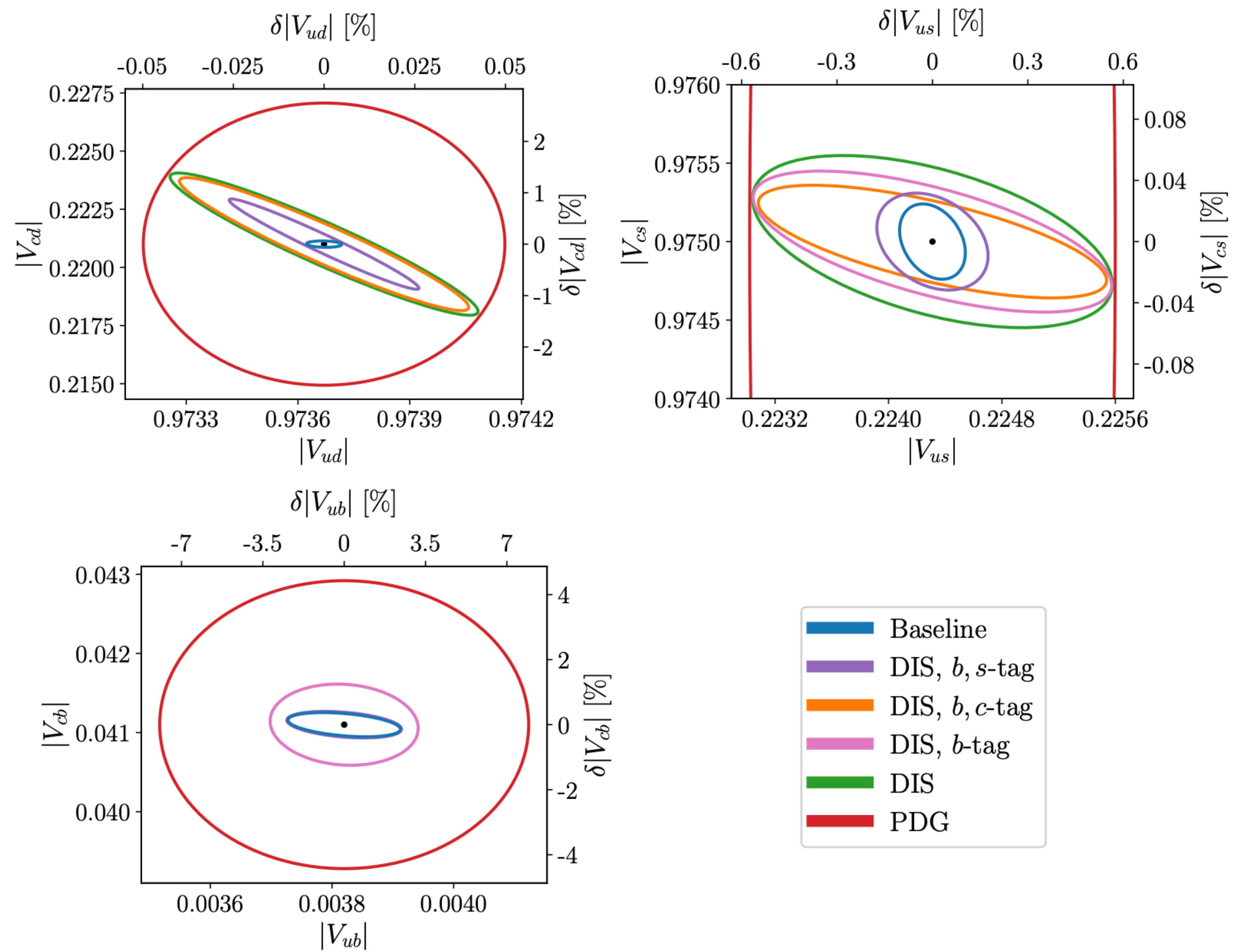
Muon-collider neutrinos add a powerful new precision program to the MuC physics case.

Backup Slides



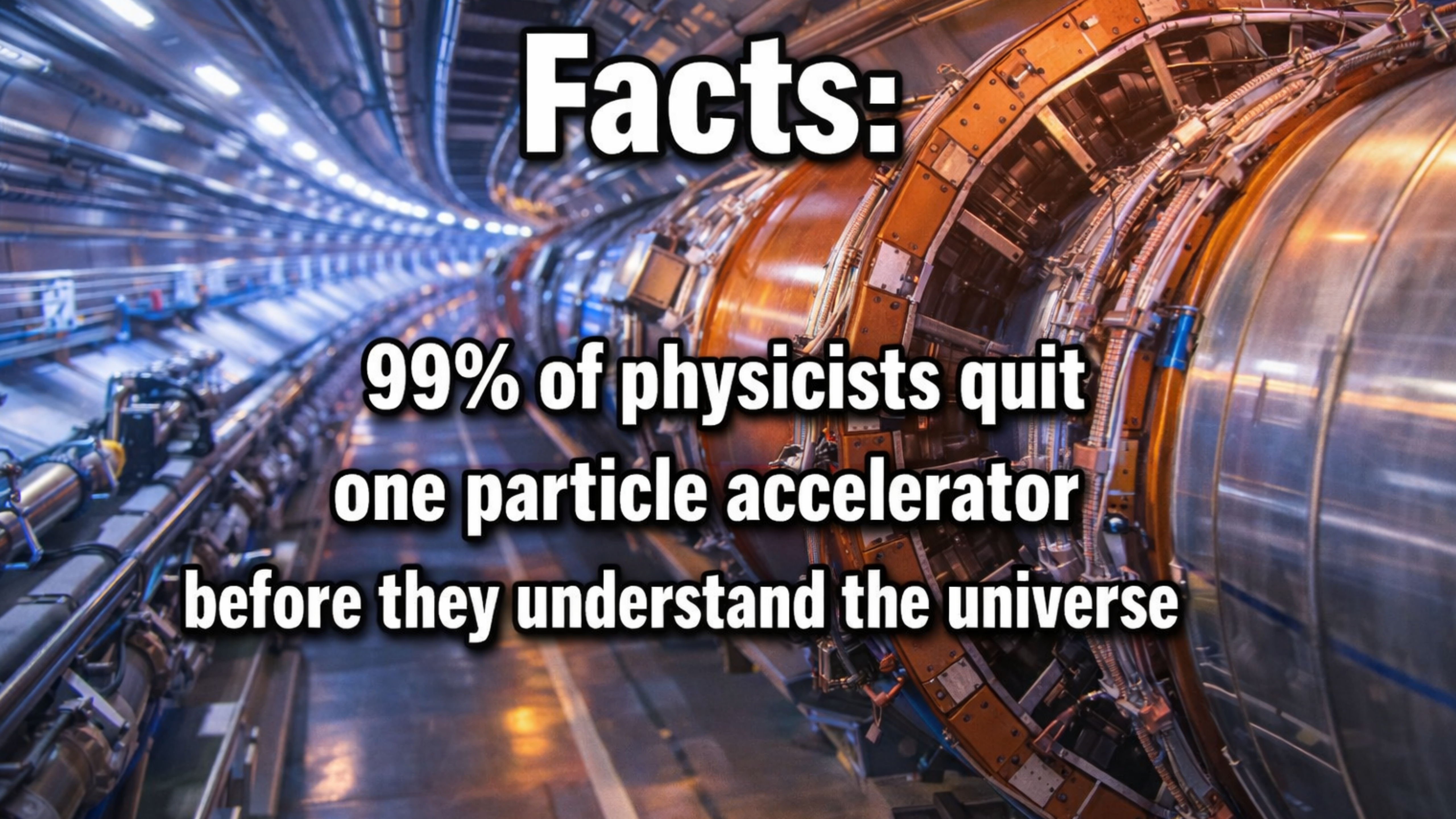






Relative Precision [%]	Proton	Neutron	Isoscalar	Silicon	Tungsten
$ V_{ud} $	0.0034	0.0032	0.0030	0.0030	0.0032
$ V_{us} $	0.049	0.10	0.043	0.051	0.067
$ V_{cd} $	0.059	0.039	0.035	0.036	0.041
$ V_{cs} $	0.016	0.018	0.010	0.011	0.016
$ V_{cb} $	0.24	0.25	0.24	0.24	0.24
$ V_{ub} $	1.3	2.1	1.5	1.5	1.6

Table 9. Relative global precision on the absolute values of CKM elements at the 10 TeV ν MuC and 1500 ab^{-1} of integrated luminosity for different target nucleons.

A photograph of a particle accelerator tunnel, showing the complex machinery and the long, curved path of the accelerator. The image is used as a background for the text.

Facts:

**99% of physicists quit
one particle accelerator
before they understand the universe**